MIL-I-25645D(USAF)
25 September 1970
Superseding
MIL-I-25645C(USAF)
27 September 1963

MILITARY SPECIFICATION

INDICATOR, LIQUID OXYGEN QUANTITY, CAPACITANCE TYPE, GENERAL SPECIFICATION FOR

1. SCOPE

1.1 This specification covers the general requirements for capacitance-type liquid oxygen quantity indicators in the following configurations:

a. Single indicator

b. Master indicator and repeater indicator.

2. APPLICABLE DOCUMENTS

* 2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

QQ-P-416	Plating, Cadmium (Electrodeposited)
QQ-Z-325	Zinc Coating, Electrodeposited, Requirements for
PPP-B-636	Box, Fiberboard

Military

MIL-P-116	Preservation, Methods of
MIL-E-5400	Electronic Equipment, Airborne, General Specification for
MIL-S-7742	Screw Threads, Standard, Optimum Selected Series: General Specification for
MIL-A-8625	Anodic Coatings, for Aluminum and Aluminum Alloys
MIL-L-25142	Luminescent Material, Fluorescent

FSC 6680

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STANDARDS

Military

MIL-STD-100	Engineering Drawing Practices
MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of US Military Property
MIL-STD-143	Standards and Specifications, Order of Precedence for the Selection of
MIL-STD-461	Electromagnetic Interference Characteristics Requirements for Equipment
MIL-STD-462	Electromagnetic Interference Characteristics, Measurement of
MIL-STD-721	Definition of Effectiveness Terms for Reliability, Maintainability, Human Factors, and Safety
MIL-STD-781	Reliability Tests: Exponential Distribution
MIL-STD-838	Lubrication of Military Equipment
MIL-STD-889	Dissimilar Metals
MIL-STD-1186	Cushioning, Anchoring, Bracing, Blocking and Waterproofing; with Appropriate Test Methods
MS33585	Pointers, Dial, Standard Design of Aircraft Instrument
MS33639	Cases, Instrument, Clamp-Mounted, Aircraft

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

* 2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials

ASTM B117-1964 Salt Spray (Fog) Testing

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

3. REQUIREMENTS

* 3.1 Type of testing. The type of testing shall be qualification or preproduction testing, or both, as specified in the detail specification.

* 3.2 Components. The indicators shall include a null balancing capacitance bridge, a transistor-type amplifier, a servo motor, and an indicating unit. Repeaters shall include a servo motor or synchro motor and indicator and may

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include the amplifier or amplifier and capacitance bridge, at the option of the contractor. Necessary circuitry shall be provided for connecting other components of the indicator system including the sensing element, press-to-test switch, and low-level-warning signal. Indicators and, where required, repeaters shall consist of the following components:

a. Case

b. Indicator dial and pointer

c. Cover glass

d. Calibration adjustments

e. Internal mechanism.

3.3 Selection of specifications and standards. Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.4 Materials

* 3.4.1 Protective treatment. When materials that are subject to deterioration when exposed to climatic and environmental conditions likely to occur during service usage are used in indicators, the materials shall be protected against such deterioration in a manner that will in no way prevent compliance with the performance requirements of this specification. The use of any protective coating that will crack, chip, or scale with age or extremes of climatic and environmental conditions shall be avoided.

3.4.2 Fungus-proof materials. Materials that are not nutrients for fungi shall be used to the greatest extent practicable. Where materials that are nutrients for fungi must be used, such materials shall be treated with a fungicidal agent acceptable to the procuring activity. However, fungicidal treatment will not be necessary if the materials will be used in a hermetically sealed inclosure.

3.4.3 Metals. Metals shall be of the corrosion-resistant type or shall be suitably treated to resist corrosion due to fuels, salt spray, oxygen, or atmospheric conditions likely to be met in storage or normal service.

* 3.4.3.1 Dissimilar metals. Unless suitably protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in MIL-STD-889.

3.4.4 Nonmagnetic materials. Except where magnetic materials are essential, nonmagnetic materials shall be used for all parts.

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3.4.5 Fluorescent material. Fluorescent material used on the indicator dials shall conform to MIL-L-25142.

* 3.5 Design and construction. The indicator shall be designed to operate in conjunction with a liquid oxygen converter having a capacitance-type sensing element immersed in the liquid oxygen. The sensing element is in a vertical position so that the liquid oxygen is the dielectric material below the liquid level and gaseous oxygen is the dielectric material above the liquid level. As the level of the oxygen in the converter changes, the electrical capacitance of the sensing element changes. This varying capacitance is connected to the bridge circuit of the indicator where a corresponding current is developed. This current is amplified and applied to a servo loop circuit which provides a null voltage which, when applied to the bridge circuit, establishes quiescence. The indicator pointer is mechanically driven by the servo motor.

* 3.5.1 Capacitance. For design purposes, the dielectric constant of liquid oxygen shall be taken as 1.483. Capacitance values for indicator readings shall be calculated as follows:

$$C = C_o + 5.8V$$

where:

C = Total capacitance, micromicrofarad (uuf)
C_o = Capacitance for zero indication, uuf
V = Volume - dial reading

The capacitance for zero indication, C_o, shall be as specified in the applicable detail specification.

3.5.2 General electrical requirements

3.5.2.1 The indicator shall be designed to operate from 115 volts, 400 hertz, single-phase power.

3.5.2.2 Electronic parts. Electronic parts and the application thereof shall be in accordance with MIL-E-5400.

3.5.2.3 Vacuum tubes and transistors. Vacuum tubes shall not be used in the electronic circuits of indicators or repeaters. Transistors used shall be subject to approval of the procuring activity.

3.5.2.4 Electrical circuits. The electrical circuits shall be designed so that the potentials will not create a fire hazard in the aircraft. The design

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of the circuits shall be such that the current in the leads running to the sensing element will in no case exceed 0.2 ampere. This requirement shall apply to both the normal operating conditions and to any combination of the following conditions:

a. Short circuiting of the sensing element leads

b. Short circuiting of any capacitor or combination of capacitors in the bridge circuit

c. All switches (and switch connections) and all adjustments set to any desired position.

3.5.2.5 Integral lighting. When specified in the detail specification, integral lighting shall be provided.

3.5.2.6 Auxiliary circuits. All single and master indicators shall have the necessary circuitry and connections for a low-level-warning indicator. Unless otherwise specified in the detail specification, provisions shall also be made for connection of an external press-to-test switch.

3.5.2.7 Electrical connectors. Electrical connectors shall be as specified in the detail specification.

3.5.3 Hermetic sealing. The indicator unit or units shall be hermetically sealed. The cases shall be filled with helium or a mixture of nitrogen and helium having at least 10 percent helium. The gas shall be under a pressure slightly above atmospheric pressure.

* 3.6 Performance. Unless otherwise specified, application of performance requirements to single units, master units, and repeater units operating from a master unit shall be as specified in the detail specification.

* 3.6.1 Accuracy - room temperature. At room temperature, the error for the single indicator, master indicator, and repeater units operating from the master indicator shall not exceed the value specified in the detail specification.

* 3.6.2 Speed of response. The pointer shall move from empty indication to full indication and from full indication to empty indication in not more than 15 seconds (for each direction) at room temperature. The requirement shall be met by the single indicator and master indicator. Repeater indicators operating from the master indicator shall obtain the same reading as the master (within limits specified in the detail specification) within 20 seconds regardless of pointer location at the start of test.

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3.6.3 Press-to-test circuit. Unless otherwise specified in the detail specification, circuitry shall be provided within the single unit or master unit so that switch operation (grounding the press-to-test pin) will cause the indicator pointer to move to the zero indication or below, and release of the switch will allow the pointer to return to its original position.

3.6.4 Low-level-warning switch. The low-level-warning switch shall close at 10 ±1 percent of full-scale reading (downward travel) of the indicator and shall remain closed down to zero reading. The switch shall carry resistive current of at least 2 amperes at 28 volts. This requirement shall apply to single units and master units.

3.6.5 Dead spot. Unless otherwise specified in the detail specification, the capacitance range required to produce movement of the indicator pointer shall not exceed 1.0 percent of full-scale indication. This requirement shall apply to single units, master units, and repeater units.

3.6.6 Sealing. Indicators shall be hermetically sealed so that leakage will be not in excess of 10^{-6} milliliters per second.

3.6.7 Low temperature operation. At a temperature of -65°F, the indicator error, with reference to the room-temperature readings, shall not exceed the value specified in the detail specification. There shall be no fogging of the cover glass while the indicator is operating at this temperature.

3.6.8 High temperature operation. At a temperature of 160°F, the indicator error, with reference to the room-temperature readings, shall not exceed the value specified in the detail specification.

* 3.6.9 Vibration. The indicator error, with reference to room temperature readings, shall not exceed the value specified in the detail specification when the indicator is subjected to a linear double amplitude or a circle diameter of 0.018 to 0.020 inch in a plane inclined 45° to the horizontal, and the frequency of vibration is varied regularly from 5 to 50 hertz and back to 5 hertz, the rate of change in frequency being constant.

* 3.6.10 Pointer oscillation vibration. The double amplitude of the pointer oscillations shall not exceed 3° when the indicator is subjected to vibration with a linear double amplitude or a circle diameter of 0.009 to 0.011 inch in a plane inclined 45° to the horizontal and the frequency of vibration is varied regularly from 5 to 50 hertz and back to 5 hertz, the rate of change in frequency being constant. This requirement shall apply to single units, master units, and repeater units.

3.6.11 Pointer overshoot. In approaching an indicator reading, the pointer shall reach its final stable position within 2 seconds. This requirement shall apply to single units, master units, and repeater units.

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3.6.12 Thermal shock. The indicator shall not be damaged as a result of eight cycles of thermal shock from 180° to 40°F.

* 3.6.13 Fogging. There shall be no evidence of moisture or oil fogging of the cover glass when the indicator is energized and maintained at a temperature of 160° ±2°F for a minimum period of 1 hour and then the face of the cover glass is rubbed with an ice cube for a period of 1 to 2 minutes.

* 3.6.14 Power supply variation. Unless otherwise specified in the detail specification, the indicator error, with reference to the room-temperature readings, shall not exceed the value specified in the detail specification when the indicator is subjected to variations of supply voltage from 105 to 125 volts and of frequency from 360 to 440 hertz.

3.6.15 Low temperature storage and operation. After being stored at a temperature of -85°F and then warmed to -65°F, the indicator error, with reference to the room-temperature readings, shall not exceed the value specified in the detail specification.

3.6.16 High temperature storage and operation. After being stored at a temperature of 160°F, the indicator error, with reference to the room-temperature readings, shall not exceed the value specified in the detail specification.

3.6.17 Low temperature response. At a temperature of -65°F, the indicator shall move from empty to full indication in not more than 45 seconds. This requirement shall apply to single units, master units, and repeater units.

* 3.6.18 Magnetic effect. The indicator shall not cause a compass deflection in excess of 5° when the indicator is positioned 5-1/2 inches from, and magnetically east or west of, the center of a compass and the horizontal magnetic field intensity at the indicator is 0.17 to 0.19 gauss.

3.6.19 Cycling. The indicator shall operate at least 5,000 cycles between zero and full indication without adverse effect on the indicator. This requirement shall apply to single units, master units, and repeater units.

* 3.6.20 Salt spray. The indicator shall operate satisfactorily after exposure to salt spray. This requirement shall apply to single units, master units, and repeater units.

3.6.21 Calibration adjustments. Calibration adjustments shall conform to the detail specification. Calibration adjustments shall be marked EMPTY and FULL, as applicable.

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* 3.6.22 Reliability. The indicator or indicator set, as applicable, shall have a specified mean cycles between failures (MCBF) of 2,000 cycles and a minimum acceptable MCBF of 1,333 cycles demonstrated at the 90 percent level of confidence. (This provides a specified reliability of .9995 for a one-cycle mission.)

3.6.22.1 Longevity. The longevity of the indicator or indicator set, as applicable, shall be not less than 7,000 cycles before wearout failures occur or the equipment consistently fails to meet the specified MCBF.

3.6.23 Maintainability. No corrective maintenance task time, as defined in MIL-STD-721, shall exceed 1 hour of maintenance downtime.

* 3.6.24 Electromagnetic interference. Indicators and repeaters shall meet the electromagnetic interference requirements of MIL-STD-461, except that the frequency range shall be 0.15 to 150 megahertz.

3.7 Details of components

* 3.7.1 Cases. Cases for indicators and repeaters shall conform to MS33639 for the 2-inch nominal size instrument case. The case for all units shall provide a hermetically sealed enclosure for the internal mechanism. The calibration adjustments need not be enclosed within the hermetically sealed part of the indicator case, but they shall be sealed in such a manner that they will withstand all environmental tests specified herein. The cases shall be designed so that the internal mechanism can be removed from the case, replaced, and the case resealed without the use of special tools or fixtures.

3.7.2 Indicator dials and pointers. Indicator dials shall be securely fastened to the frames of the indicator mechanisms in such a manner that they will not turn or loosen when the instruments are vibrated. All markings shall be durable to withstand usage encountered in service. Indicator dials shall be as specified in the detail specification. Dial scales shall be evenly graduated between zero and full-scale markings, which shall comprise a $180^{\circ} \pm 1^{\circ}$ sector. Pointers shall conform to MS33585-8. Pointers and dials shall be firmly attached to the associated mechanism.

3.7.3 Cover glass. The cover glass shall be suitable for hermetic sealing and shall be clear, flat, and free from flaws which would interfere with the normal reading of the indicator. The cover glass shall be as thin as practicable. The distance from the front surface of the bezel and the outside surface of the cover glass shall not exceed 0.062 inch.

3.7.4 Calibration adjustments. Calibration adjustments for the single indicator and the master indicator shall be provided to compensate for length of cable and other factors in installation. These adjustments shall be

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accessible at the back of the cases for field adjustment. Provision shall be made so that the adjustments will not be disturbed by shock or vibration. These requirements apply to single units and to master units. If necessary to meet the performance requirements, calibration adjustments may be used in repeater units.

3.7.5 Internal mechanisms. The internal mechanism shall consist of all circuitry and components necessary to meet the performance requirements specified herein.

* 3.8 Part numbering of interchangeable parts. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The item identification and part number requirements of MIL-STD-100 shall govern the manufacturer's part numbers and changes thereto.

3.9 Weight. The maximum weight of each unit shall conform to the detail specification.

3.10 Finishes and protective coatings

3.10.1 Aluminum alloy parts. Where practicable, aluminum alloy parts shall be anodized in accordance with MIL-A-8625.

3.10.2 Steel parts. Steel parts shall be zinc or cadmium plated in accordance with type II or III of QQ-Z-325 or type II or III of QQ-P-416, respectively.

3.11 Screw threads. The threads of all machine screws 0.060 inch or larger in diameter shall conform to MIL-S-7742.

3.12 Lubrication. Lubrication shall be accomplished in accordance with MIL-STD-838.

3.13 Identification of product. Indicators shall be marked for identification in accordance with MIL-STD-130 and the detail specification.

3.14 Workmanship. The indicator, including all parts and accessories, shall be constructed and finished in a thoroughly workmanlike manner. Particular attention shall be given to neatness and thoroughness of soldering, wiring, marking of parts and assemblies, welding and brazing, painting, riveting, machine-screw assemblies, and freedom of parts from burrs and sharp edges.

3.14.1 Tightness of screws and bolts. Screws and bolts shall be tight so that they cannot be appreciably tightened further without damage to the screw or bolt.

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3.14.2 Riveting. Riveting operations shall be carefully performed to insure that the rivets are tight and satisfactorily headed.

3.14.3 Gear assemblies. Gear assemblies shall be properly aligned and meshed and shall be operable without interference, tight spots, loose spots, or other irregularities. Where required, gear assemblies shall be free from backlash.

3.14.4 Cleaning. After final assembly, the indicator shall be thoroughly cleaned of loose, spattered, or excess solder, metal chips, and any other foreign material. Burrs and sharp edges, as well as resin flash that might crumble, shall be removed.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.2 Classification of tests. The inspection and testing of indicators shall be classified as follows:

a. Qualification or preproduction tests, or both, as specified in the detail specification

b. Quality conformance tests.

4.3 Test conditions

4.3.1 Atmospheric conditions. Unless otherwise specified, all tests required by this specification shall be made at an atmospheric pressure of approximately 29.92 inches of mercury, a temperature of approximately 77°F, and a relative humidity of 80 percent or less. Where tests are made with atmospheric pressure or temperature substantially different from these values, proper allowance shall be made for the change in instrument reading.

4.3.2 Position. Unless otherwise specified, the indicator shall be tested with the dial in the normal operating position.

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* 4.3.3 Test power. Unless otherwise specified herein, the indicators shall be tested with an electrical power source of 115 volts, 400 hertz, single-phase current. The voltage and frequency tolerance shall be ±2 percent.

4.3.4 Master test instrument. The master test instrument shall be a precision variable capacitor having an accuracy of within ±0.2 uuf or ±0.2 percent, whichever is the greater.

4.3.5 Components. Where applicable, indicators and repeaters shall be tested together.

4.3.6 Test connector. Unless otherwise specified, all functional tests of the indicator or indicator set shall be conducted utilizing the mating connector specified in the detail specification.

* 4.4 Quality conformance tests. Quality conformance tests shall consist of:

a. Individual tests

b. Sampling plans and tests.

4.4.1 Individual tests. Each indicator shall be subjected to the following tests as described under 4.5:

a. Examination of product

b. Accuracy - room temperature

c. Speed of response

d. Press-to-test circuit

e. Low-level-warning switch

f. Dead spot

g. Sealing.

* 4.4.2 Sampling plans and tests

* 4.4.2.1 Sampling plan A. A sample of indicators shall be randomly selected from each lot in accordance with MIL-STD-105, special inspection level S-4, acceptable quality level (AQL) 2.5 percent, and subjected to the following tests as described under 4.5:

a. Low temperature operation

b. High temperature operation

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c. Vibration

d. Pointer oscillation-vibration

e. Pointer overshoot

f. Thermal shock

g. Fogging.

* 4.4.2.2 Sampling plan B. Indicator samples shall be subjected to the reliability test described in 4.5.22. Sampling for the reliability tests shall be in accordance with MIL-STD-781. The level of testing shall be as specified in 4.5.22. These tests shall be required for each lot.

* 4.4.2.3 Rejection and retest. Rejection and retest of lots shall be in accordance with MIL-STD-105 for sampling plan A and in accordance with MIL-STD-781 for sampling plan B.

4.5 Test methods

4.5.1 Examination of product. The indicators shall be carefully examined to determine conformance to the requirements of this specification with respect to material, workmanship, weight, design, construction, and marking.

* 4.5.2 Accuracy - room temperature. The indicator or master indicator shall be connected to a master test instrument. The test instrument shall be set to a capacitance value corresponding to the empty converter, and adjustments shall be made on the indicator to place the pointer on the empty position. The same procedure shall be followed to adjust the instrument at the full position. When a repeater is used with a master indicator, it may or may not include an adjustment mechanism. No further adjustments shall be made during the remainder of the test. The test instrument shall be varied so that the indicator or master indicator indicates at each major graduation, and the capacitance input to the indicator or master indicator shall be noted. Capacitance readings shall be taken at each of these points, and these readings shall not vary from capacitance calculated from the formula in 3.5.1 by more than the values specified in the detail specification. When a repeater indicator is used, it shall be tested in conjunction with the master unit, and the accuracy shall be within the limits of the detail specification. These capacitance readings shall be used as a reference for determining subsequent scale errors.

* 4.5.3 Speed of response. The single indicator or master indicator shall be connected to a master test instrument. The master test instrument shall be adjusted so that the indicator pointer travels from zero to full indication and back. Travel in either direction shall require not more than 15 seconds

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at room temperature. Repeater indicators operating from the master indicator shall meet the specified reading within 20 seconds regardless of pointer location at the start of test.

4.5.4 Press-to-test circuit. The circuitry within the indicator shall be tested by grounding the press-to-test pin and checking to see that the indicator pointer moves downward to zero indication or below when the button is pressed and returns when the button is released.

* 4.5.5 Low-level-warning switch. This test may be conducted in conjunction with either of the two preceding tests by connecting a light or other suitable device in the system. The light or other device shall remain actuated throughout the range of 0 to 10 $\pm$1 percent of full scale of the indicator.

4.5.6 Dead spot. The input capacitance of the indicator shall be slowly varied by means of a master test instrument, first in one direction and then in the other. The capacitance change required to rotate the indicator pointer shall be noted and shall be computed as a percentage of the full-scale indication for which the indicator is calibrated. This percentage is defined as the dead spot and shall not exceed 1.0 percent.

4.5.7 Sealing. Indicators shall be tested for adequacy of the hermetic seals using a helium leak detector. The leakage rate shall not exceed 10^{-6} milliliters per second.

4.5.8 Low temperature operation. The indicator shall be subjected to a temperature of -65° $\pm$2°F for a period of at least 4 hours. The master test instrument shall be kept at room temperature. At the end of the 4-hour period while the indicator is still at the low temperature, and after the indicator has been energized for at least 6 minutes, scale readings shall be determined for each major graduation by means of the master test instrument. The scale error shall not exceed the value specified in the detail specification. There shall be no fogging of the cover glass during this test.

4.5.9 High temperature operation. The indicator shall be subjected to a temperature of 160° $\pm$2°F for at least 4 hours. The master test instrument shall be kept at room temperature. At the end of the 4-hour period, while the indicator is still at the high temperature, scale readings shall be determined for each major graduation by means of the master test instrument. The scale error shall not exceed the value specified in the detail specification.

* 4.5.10 Vibration. The indicator shall be electrically energized and set to read approximately midscale by means of the master test instrument. The indicator shall be subjected to vibration with a linear double amplitude or a circle diameter of 0.018 to 0.020 inch in a plane inclined 45° to the horizontal for a period of 3 hours. The frequency of vibration shall vary regularly from 5 to 50 hertz and back to 5 hertz, the rate of change in frequency being

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constant. Following the 3-hour period, a reading shall be taken for each major graduation using the master test instrument. The scale error shall not exceed the value specified in the detail specification.

* 4.5.11 Pointer oscillation vibration. The indicator shall be subjected to vibration with a linear double amplitude or a circle diameter of 0.009 to 0.011 inch in a plane inclined 45° to the horizontal. The frequency of vibration shall vary regularly from 5 to 50 hertz and back to 5 hertz, the rate of change in frequency being constant. The maximum double amplitude of the pointer oscillation shall not exceed 3°. The test shall be conducted at a reading of zero and at a minimum of two additional points.

4.5.12 Pointer overshoot. The indicator shall be tested at each major graduation with the master test instrument. In approaching a reading, the pointer shall reach its final stable position within 2 seconds.

* 4.5.13 Thermal shock. The indicator shall be alternately subjected to a total of eight cycles of immersion in tap water maintained at 180° ±2°F and 40° ±2°F. The length of time for each immersion shall be 30 minutes, and not more than 5 seconds shall elapse between immersions. No evidence of moisture penetration or damage to the indicator shall result from this test. Changes in temperature from the high to the low extremes during this test shall not cause fogging of the internal surface of the cover glass. Following this test, the indicator shall be rechecked for leakage.

4.5.14 Fogging. The indicator shall be energized and maintained in a 160° ±2°F controlled ambient for a minimum period of 1 hour. After this period and with the indicator maintained at this temperature, an ice cube shall be rubbed on the indicator cover glass for a period of from 1 to 2 minutes. The cover glass shall then be wiped dry (compressed air shall not be used) and the indicator inspected for evidence of water or oil fog. Evidence of fogging shall be cause for rejection.

* 4.5.15 Power supply variation. The indicator shall be tested at the three different positions of the pointer: approximately zero, midscale, and full. For this testing the combinations of power supply, frequency, and voltage shall be as follows:

Voltage	Frequency
115 ±1.0 volts	400 ±5 hertz
105 ±1.0 volts	360 ±5 hertz (unless otherwise specified in the detail specification)
125 ±1.0 volts	440 ±5 hertz (unless otherwise specified in the detail specification)

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The change in indication from the nominal power source (115 volts, 400 hertz) shall not exceed the value specified in the detail specification.

4.5.16 Low temperature storage and operation. The indicator shall be kept at a temperature of -85° ±2°F for a period of 18 hours. The temperature shall then be raised to -65° ±2°F and maintained for a period of 6 hours. During the last 2 hours of the 6-hour period, the ambient pressure shall be reduced to 0.82 inch of mercury absolute, or less. At the completion of the 24-hour period and while still at the temperature and the pressure specified, and after the indicator has been energized for at least 6 minutes, the indicator shall be tested to determine that the scale error does not exceed the value specified in the detail specification.

4.5.17 High temperature storage and operation. The indicator shall be subjected to a temperature of 160° ±2°F for a period of at least 24 hours. While still at this temperature and at the end of the 24-hour period, readings shall be taken at each major graduation. The scale error shall not exceed the value specified in the detail specification.

4.5.18 Low temperature response. The indicator shall be maintained at a temperature of -65° ±2°F for at least 2 hours. While still at this temperature, the time required for the pointer to travel from zero to full shall not exceed 45 seconds.

4.5.19 Magnetic effect. The indicator, while not being operated, shall be rotated in a vertical plane about a short bar magnet compass with the nearest part of the indicator 5-1/2 inches from, and magnetically east or west of, the center of the compass. Starting directly under the compass, the indicator shall be held in positions 0, 45, 90, 135, 180, 225, 270, and 315 degrees from the initial position. At each of these positions the indicator shall be rotated on its own horizontal axis until it is in its normal upright position. The horizontal magnetic field intensity shall be 0.17 to 0.19 gauss. The deflection of the compass at any of the specified positions shall not exceed 5°. The test shall be repeated with the indicator at the rated voltage.

* 4.5.20 Salt spray. The indicator shall be subjected to salt spray in accordance with ASTM test method B117 for a period of 50 hours. Readings shall be taken at each major graduation, and the scale error shall not exceed the value specified in the detail specification. After this test any evidence of corrosion or any other deterioration that might affect subsequent operation of the indicator shall be cause for rejection.

4.5.21 Calibration adjustments. The input capacitance shall be set as specified in the detail specification. By means of the calibration adjustments, the pointer shall be adjusted to the proper indications (empty and full). Failure of the pointer to adjust to these indications shall be cause for rejection.

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* 4.5.22 Reliability test level. The reliability test level shall consist of cycling of the indicator or indicator set as follows: The indicator or indicator set shall be connected to the master test instrument and the capacitance continuously varied so that the pointer moves from zero to full and returns to zero within 5 ±1 minutes. The cycling period shall be conducted in 12 cycle intervals. After each interval the indicator shall be de-energized electrically a minimum of 10 minutes between cycling intervals. A failure shall be identified as any malfunction which causes excessive scale error of the indicator or indicator set. No parts shall be replaced as preventive maintenance during reliability testing. The indicator or indicator set shall be checked at least once every 24 hours during reliability testing to determine if a failure has occurred. Empty and full calibration adjustments of the indicator shall not be made during reliability testing, unless necessary to bring the scale error measurements within the specified tolerance, at which time appropriate records shall be made in the reliability test data. Acceptance for the reliability test level shall be judged on the basis of the accept-reject criteria of test plan I of MIL-STD-781. Recording, data handling, and reporting shall be in accordance with MIL-STD-781.

* 4.5.23 Longevity. The longevity test shall consist of continuing the reliability test (4.5.22) through a longevity test in accordance with MIL-STD-781.

4.5.24 Maintainability verification. Time studies of all corrective maintenance tasks for all expected failure modes shall be conducted by simulating failures. Timing of each maintenance task shall be performed at least three times to determine average values. The maintenance tasks considered shall be only those related to field level maintenance. The time to accomplish each maintenance task shall include time for preparation, recognition, fault location, removal (based only on the installation provisions of the item), fault correction, adjustment and realignment, final test, and replacement, but shall exclude supply time and administrative time. In the case of repetitive subtasks which occur by the nature of the sequence of maintenance actions (e.g., removal), it shall not be necessary to redetermine the average time required.

* 4.5.25 Electrical interference. The indicator shall be subjected to and shall pass the applicable tests specified in MIL-STD-462.

4.6 Inspection of preparation for delivery. Sample items or packs and the inspection of the preservation, packaging, packing, and marking for shipment and storage shall be in accordance with the requirements of section 5, or the documents referenced therein.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C, as specified (see 6.2).

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5.1.1 Level A. Each indicator shall be preserved and packaged in accordance with MIL-P-116, method III, and placed in a snug-fitting container conforming to PPP-B-636, grade W6c.

5.1.2 Level C. Each indicator shall be preserved and packaged in a manner which will afford adequate protection against corrosion, deterioration, and physical damage during shipment from supply source to the first receiving activity. The supplier may use his commercial practice provided it fulfills this requirement.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Level A. Indicators preserved and packaged as specified in 5.1.1 shall be packed in weather-resistant shipping containers conforming to PPP-B-636. The grade of each shipping container, determined by the gross weight and cube, shall comply with the applicable special requirements table of the specification. Interior cushioning, blocking, bracing, and waterproofing shall be in accordance with MIL-STD-1186. Closure and strapping shall be in accordance with the appendix of PPP-B-636. Where practicable, the exterior shipping containers shall be of uniform shape and size and of minimum tare and cube.

5.2.2 Level B. Level B packing shall be the same as level A except that waterproofing will not be required and domestic shipping containers may be used.

5.2.3 Level C. Packages that require overpacking for acceptance by the carrier shall be placed in exterior-type shipping containers in a manner that will ensure safe transportation at the lowest rate to the point of delivery. Containers shall comply with the Uniform Freight Classification Rules or regulations of other carriers as applicable to the mode of transportation.

5.3 Marking. In addition to any other marking required by the order or contract, interior and exterior containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The indicators covered by this specification are intended to indicate the quantity of liquid oxygen in a liquid oxygen converter.

* 6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification

b. Levels of preservation, packaging, and packing required (see section 5).

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* 6.3 International agreement. The unit of measurement used for oxygen (liters) in this specification is the subject of international standardization agreement ASCC Air Standard 10/2. When amendment, revision, or cancellation of this specification is proposed which will affect or violate the international agreement concerned, the preparing activity will take appropriate reconciliation action through international standardization channels, including departmental standardization offices, if required.

* 6.4 The margins of this specification are marked with an asterisk to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodian:
Air Force - 11

Review activity:
Air Force - 82

User activity:
Air Force - 71

Preparing activity:
Air Force - 11

Project No. 6680-F131

☆ U. S. GOVERNMENT PRINTING OFFICE: 1971-433-695/8849

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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